LECTURE 8

LAST TIME

LECTURE 7: SUMMARY

Portfolios of options

Corporate securities

THIS TIME

LECTURE 8 OUTLINE

Binomial model

- Stock prices on the binomial tree
- Expected return and variance on the tree
- Options on the tree

THE BINOMIAL MODEL

IMPLICATIONS OF NA ALONE

So far, no assumptions on distributions of prices/returns

NA gave unique forward price (no storage cost/convenience yield)

NA gave bounds on prices on derivatives like options

How can we obtain more precise price estimates for derivatives?

SIMPLEST RANDOM MODEL

The simplest outcome is deterministic

i.e. a random variable with one possible outcome

That is how we model a bond

A "MODEL" OF A BOND



THE BOND PRICE

today tomorrow

$$\frac{1}{1+r} = B_t \qquad 1$$

THE SIMPLEST REALLY RANDOM MODEL

The simplest random model has two outcomes

We will use the following notation

$\omega \in \{d, u\}$ is the underlying state

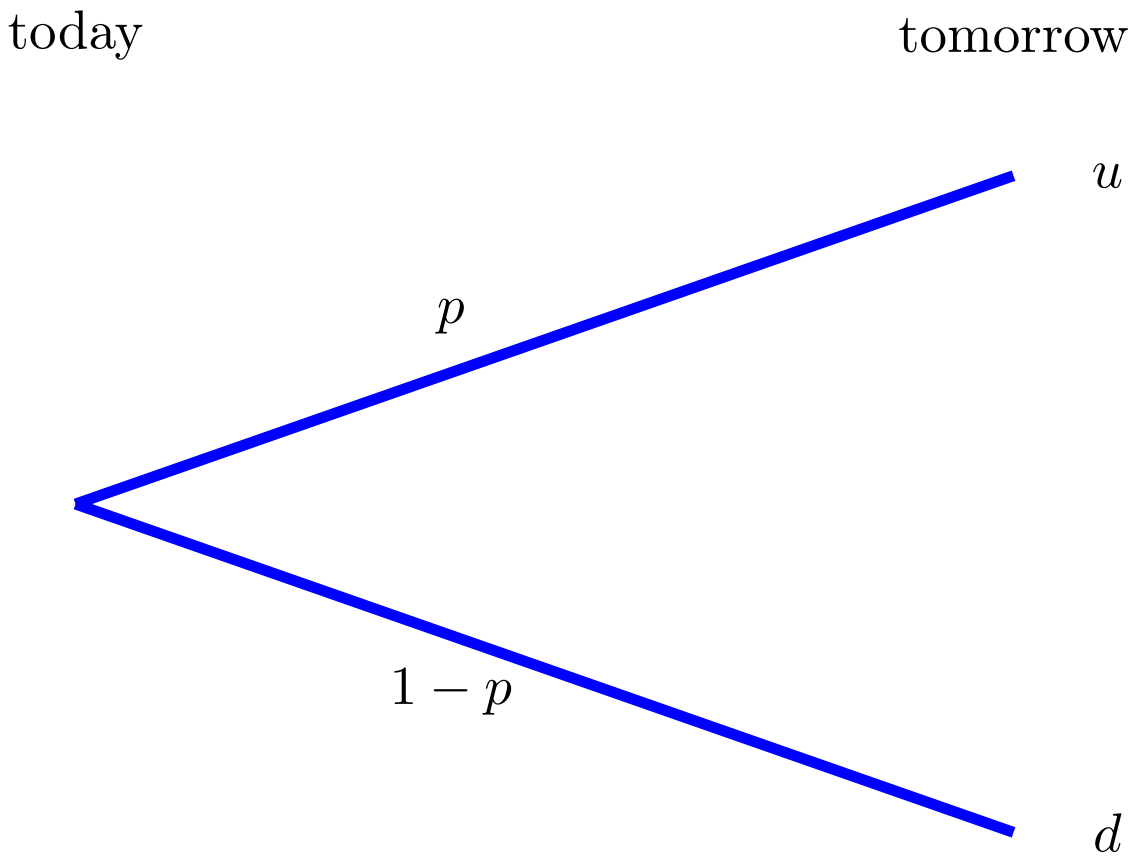
p is the probability of an "up" state

i.e. $p := \mathbb{P}[\omega = u]$

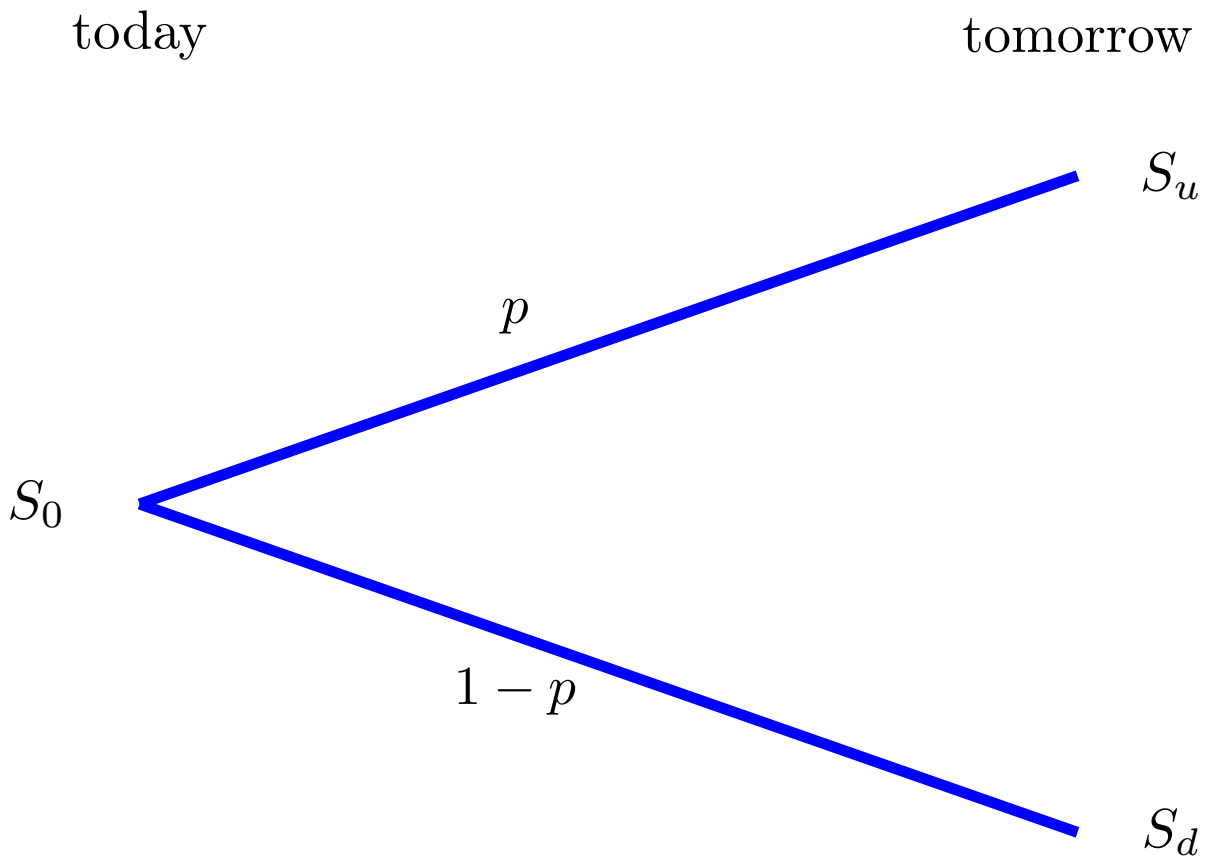
$S_\omega \equiv S_T(\omega)$ is the payoff of asset S in state ω

S_0 is the price of asset S_0 today

THE STATES TOMORROW



ASSET PAYOFF AND PRICE



EXPECTED RETURN ON THE BINOMIAL TREE

What is the expected gross return of S?

$$\mathbb{E}\left[\frac{S_T}{S_0}\right] = \frac{pS_u + (1-p)S_d}{S_0}$$

What is the variance of the return of S?

$$\begin{aligned}\text{Var}\left[\frac{S_T}{S_0}\right] &= \mathbb{E}\left[\left(\frac{S_T}{S_0} - \mathbb{E}\left[\frac{S_T}{S_0}\right]\right)^2\right] \\ &= \mathbb{E}\left[\left(\frac{S_T}{S_0}\right)^2\right] - \left(\mathbb{E}\left[\frac{S_T}{S_0}\right]\right)^2\end{aligned}$$

Volatility typically refers to square root of variance of return

BINOMIAL TREE WARM-UP EXERCISE

Consider binomial model with $p = 1/2$

There are two assets, S^1 and S^2 such that

$S_0^1 = 1$, $S_d^1 = 0$, $S_u^1 = 4$

$S_0^2 = 1$, $S_d^2 = 2$, $S_u^2 = 2$

Compute the expected returns and volatilities of S^1 and S^2

What is the risk free rate (between "today" and "tomorrow")?

Which security has higher return? Which has higher volatility?

Is that reasonable/possible given NA?

BINOMIAL TREE WARM-UP SOLUTION

For S^1:

$$\mathbb{E}\left[\frac{S_T^1}{S_0^1}\right] = \frac{1}{2}\cdot\frac{4}{1} + \frac{1}{2}\cdot\frac{0}{1} = 2$$

$$\begin{aligned}\text{Var}\left[\frac{S_T^1}{S_0^1}\right] &= \mathbb{E}\left[\left(\frac{S_T^1}{S_0^1} - \mathbb{E}\left[\frac{S_T^1}{S_0^1}\right]\right)^2\right] \\ &= \frac{1}{2}\cdot\left(\frac{4}{1} - 2\right)^2 + \frac{1}{2}\cdot\left(\frac{0}{1} - 2\right)^2 \\ &= 4\end{aligned}$$

thus, the vol. of the return of S^1 is 2

BINOMIAL TREE WARM-UP SOLUTION

For S^2:

$$\mathbb{E}\left[\frac{S_T^2}{S_0^2}\right] = \frac{1}{2}\cdot\frac{2}{1} + \frac{1}{2}\cdot\frac{2}{1} = 2$$

$$\begin{aligned}\text{Var}\left[\frac{S_T^2}{S_0^2}\right] &= \mathbb{E}\left[\left(\frac{S_T^2}{S_0^2} - \mathbb{E}\left[\frac{S_T^2}{S_0^2}\right]\right)^2\right] \\ &= \frac{1}{2}\cdot\left(\frac{2}{1} - 2\right)^2 + \frac{1}{2}\cdot\left(\frac{2}{1} - 2\right)^2 \\ &= 0\end{aligned}$$

thus, the vol. of the return of S^2 is 0

BINOMIAL TREE WARM-UP SOLUTION

The risk-free rate is 100%

NA implies risk-free rate unique and S^2 is risk free

S^1 and S^2 have the same return and S^1 has higher variance

NA does not rule this out

In the CAPM world, S^2 has beta zero

TWO ASSET BINOMIAL MODEL

We often focus on a binomial tree with two assets

risk-free bond B, $B_u = B_d$

risky stock S, $S_u > S_d$

It is often convenient to use returns instead of prices

write u for return in state u

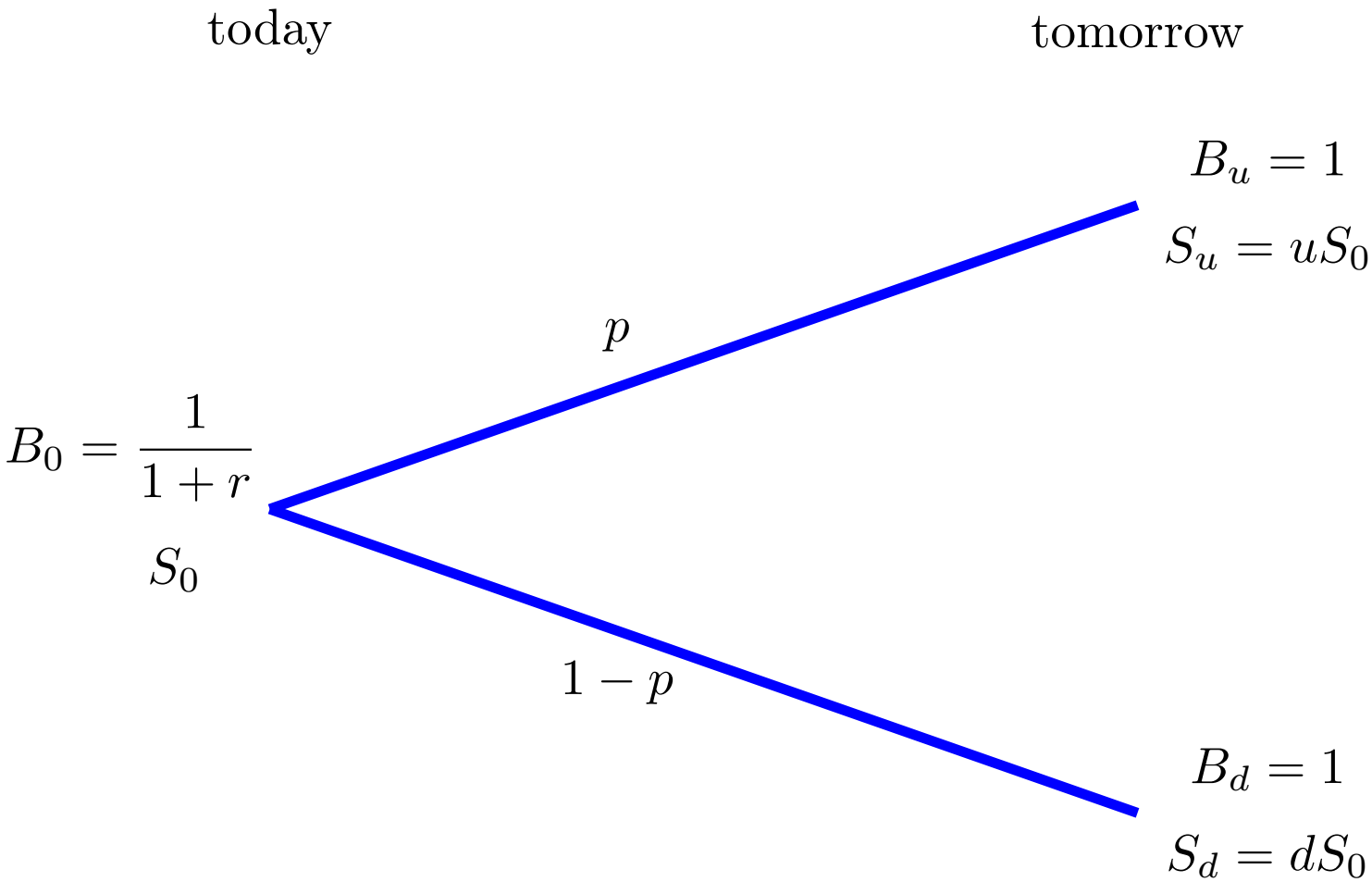
write d for return in state d

thus

$$u := \frac{S_u}{S_0} \quad \text{and} \quad d := \frac{S_d}{S_0}$$

Note: use u and d for both states (ω) and numbers (returns)

ASSET PAYOFF AND PRICE



QUICK NA PROOF PRACTICE

Show (as long as $u > d$) that NA implies

$$d \leq 1 + r \leq u$$

Proof that $d \leq 1 + r$:

Suppose $d > 1 + r$

Short one dollar of the bond and buy one dollar of the stock

$$\text{CF}_0 = 1 - 1 = 0$$

and

$$\text{CF}_d = \frac{S_d}{S_0} - (1 + r) = d - 1 - r \geq 0$$

and

$$\text{CF}_u = \frac{S_u}{S_0} - (1 + r) = u - 1 - r > d - 1 - r \geq 0$$

by assumption. Arbitrage! Thus $d \leq 1 + r$

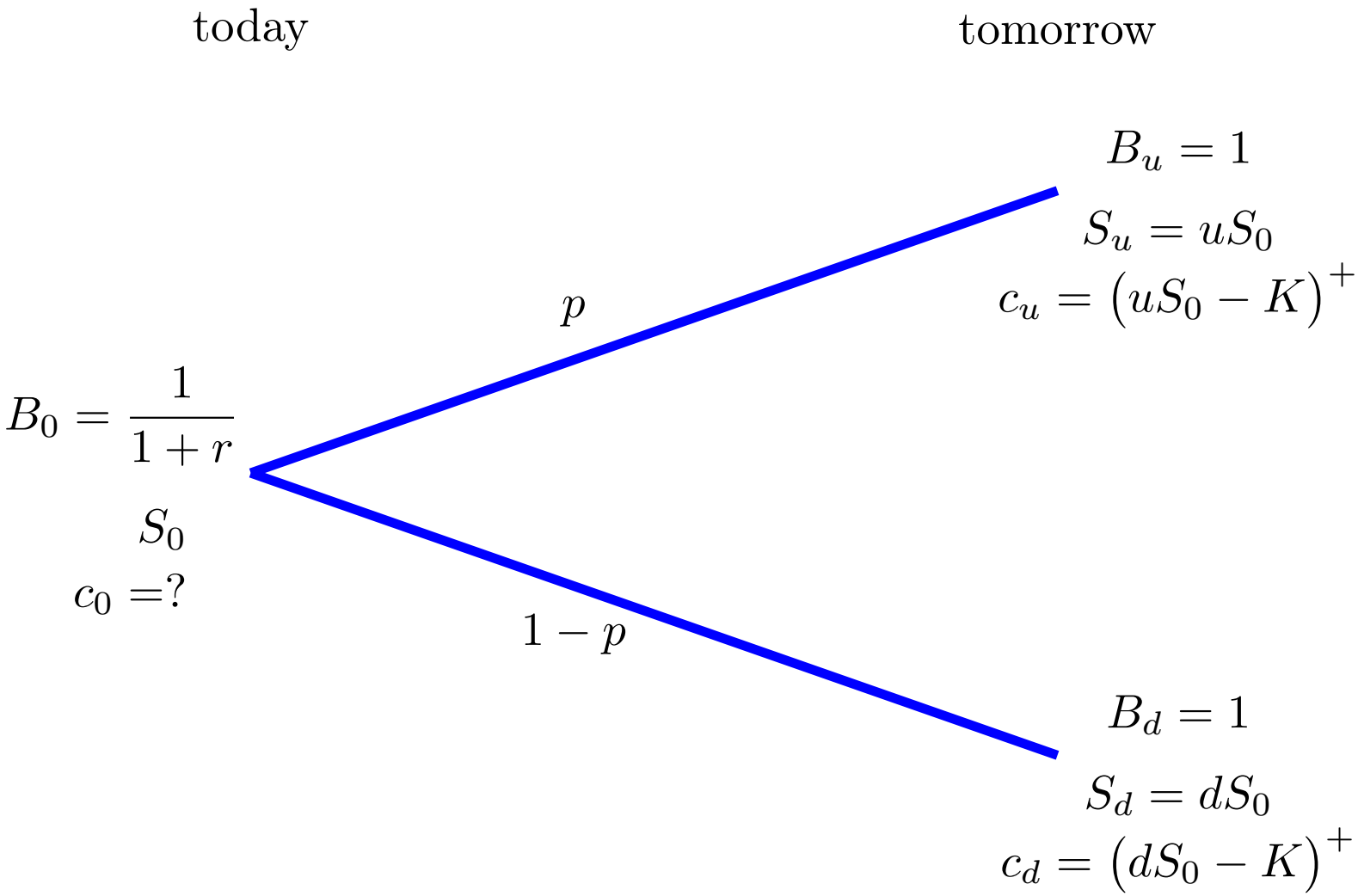
HW: show $1 + r \leq u$ by finding a contradiction if $1 + r > u$

PRICING A CALL ON THE BINOMIAL TREE

Consider a call with strike price K

Can the binomial model help to find its price, c_0?

CALL OPTION PAYOFFS



RESTRICT K

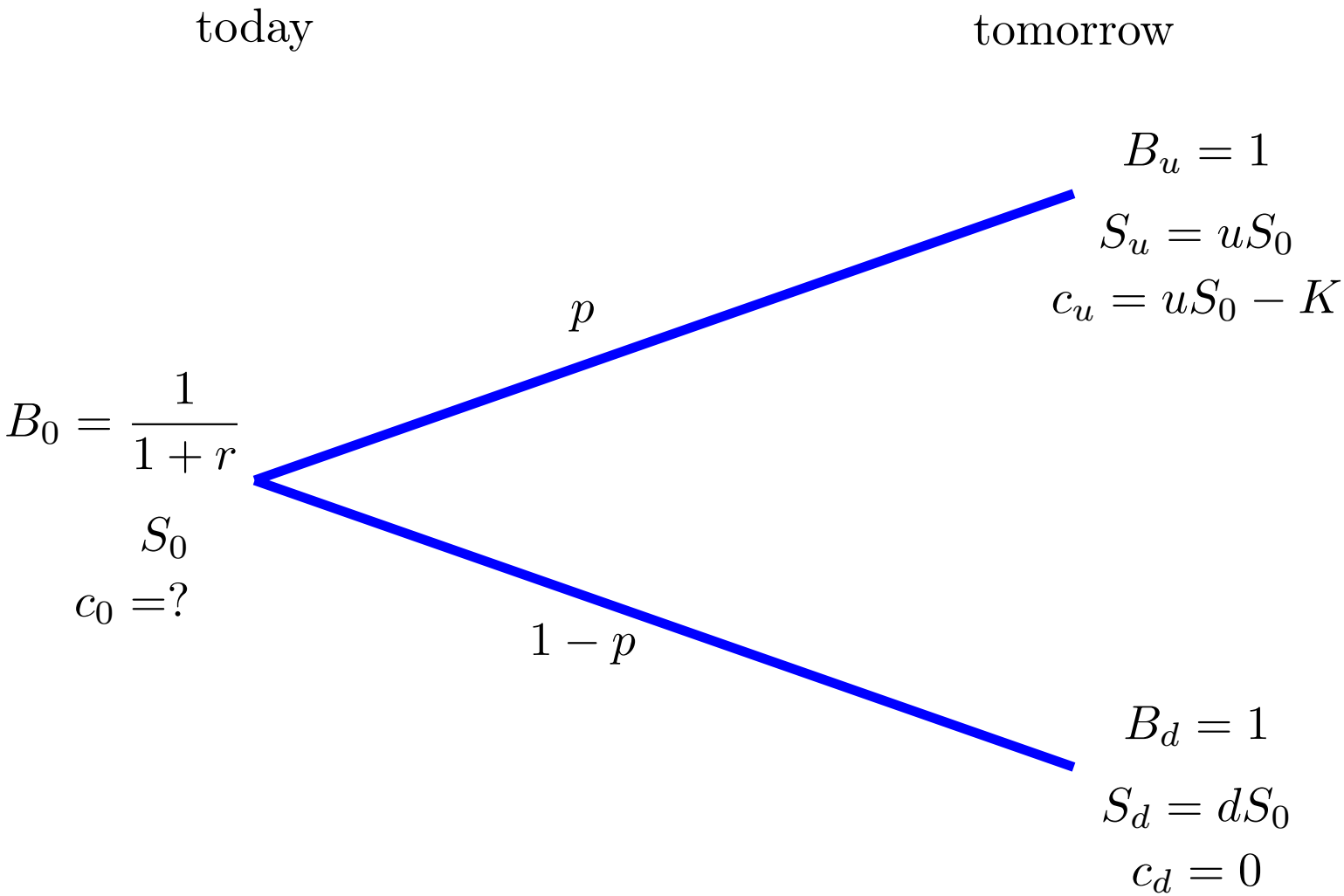
Focus on $dS_0 < K < uS_0$ w.l.o.g.

If $K > uS_0$, payoff always zero

If $K < dS_0$, payoff always $S_\omega - K$, a long forward

So $dS_0 < K < uS_0$ only case we not already studied

CALL OPTION PAYOFFS



PRICING BY REPLICATION

Want to price by replication

So far, found portfolios that had identical payoffs to instrument

- Didn't write down dependence on states
- Payoffs coincided for every state
- Could price forwards this way
- But didn't have enough structure to price options

With two states, can replicate options state by state

REPLICATING PORTFOLIO FOR A CALL

Form a portfolio with β bonds and Δ stocks

Portfolio replicates a call in every state, i.e. for $\omega = u$ and $\omega = d$:

$$c_u = \beta B_u + \Delta S_u$$

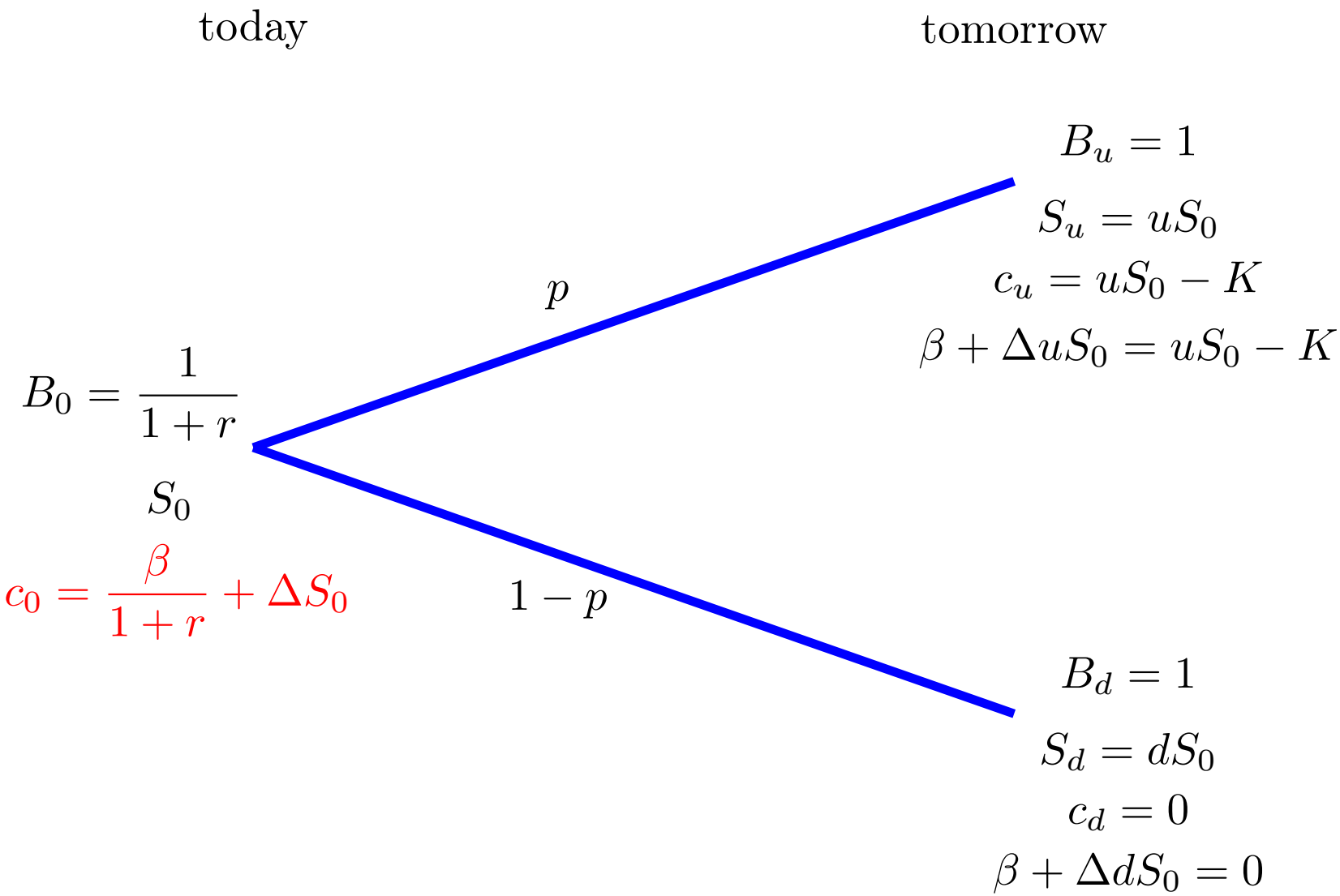
$$c_d = \beta B_d + \Delta S_d$$

If we find β and Δ, can use LoOP to price of the call:

$$c_0 = \beta B_0 + \Delta S_0$$

Is replication possible? Not always, but can try here

HEDGING PORTFOLIO



REPLICATING PORTFOLIO EQUATIONS

Given $B_\omega = 1$, $S_\omega = \omega S$ and expressions for c_u and c_d, β and Δ satisfy

$$\beta + \Delta u S_0 = u S_0 - K$$

$$\beta + \Delta d S_0 = 0$$

Solving:

$$\Delta = \frac{u S_0 - K}{(u-d) S_0}$$

$$\beta = -\frac{d\big(u S_0 - K\big)}{u - d}$$

APPLY THE LAW OF ONE PRICE

Applying the Law of One Price gives

$$\begin{aligned} c_0 &= \frac{\beta}{1+r} + \Delta S_0 \\ &= -\frac{d\big(uS_0 - K\big)}{(u-d)(1+r)} + \frac{uS_0 - K}{u-d} \end{aligned}$$

PRICING ALGORITHM

PRICING ALGORITHM

Steps to price a derivative X on binomial tree:

I. Find payoffs X_u and X_d

II. Write eqs. for replicating portfolio (β, Δ) in states u and d

III. Solve system for β and Δ

IV. Apply LoOP to find X_0

PRICE A PUT EXERCISE

Let $S_0 = 1$, $u = 2$, $d = 0$, $r = 0$, $p = 1/2$

Price a put with strike $K = 1$

PRICE A PUT SOLUTION

In state u

$$\beta + \Delta u S_0 = 0$$

or

$$\beta + 2\Delta = 0$$

In state d

$$\beta + \Delta d S_0 = K - d S_0 = 1$$

or

$$\beta = 1$$

Thus, $\Delta = -1/2$ and

$$p_0 = 1 - \frac{1}{2} = \frac{1}{2}$$

Observe that $p_0 = \mathbb{E}\left[p_T\right]$. Could you have guessed this?

RECAP

LECTURE 8: RECAP

The binomial model

Options on the binomial tree

FIN